ISTE 730 – Foundations of IoT

Automated Logistics and Access Security Gate with Monitoring and Behavioral Analysis

Vetrivel Maheswaran

1. Introduction

Industrial facilities, warehouses, and logistics centers require secure access control systems to ensure that only authorized personnel or vehicles can enter restricted areas. Traditional access systems often rely only on authentication at the point of entry and do not maintain detailed records or provide insights into usage behavior. This limits their ability to monitor security trends or detect unusual access patterns. This project is based on the concept of an Automated Logistics and Access Security Gate, where an Arduino-based system controls a physical gate using RFID authentication. When a user scans an RFID card, the system verifies whether the card is authorized and allows or denies access accordingly. While this satisfies the basic requirement of access control, the system can be improved by integrating data logging, monitoring, and simple behavioral analysis.

The motivation for this project comes from an interest in combining hardware-level control with data-driven intelligence. This work aligns with real-world applications in smart security and IoT-based monitoring systems, while also supporting learning objectives in integrating embedded systems with backend analytics and lightweight machine learning.

2. Goal

The goal of this project is to design and implement a smart access security gate system that performs real-time authentication and maintains a record of access events while providing basic monitoring and anomaly detection capabilities. The system will authenticate users using RFID cards, control gate access using a servo motor, and provide feedback through LEDs, buzzer, and display. It will log all access events with timestamps in a database and provide a simple dashboard for real-time monitoring and history tracking. Additionally, the system will perform basic behavior analysis and detect unusual access activity using a lightweight method. The final system will be a hardware-driven security gate integrated with a backend system for logging, monitoring, and basic anomaly detection.

3. Methodology

The system will be implemented using an Arduino-based hardware setup integrated with a Python backend for data processing and monitoring. The software stack uses Arduino C/C++ for microcontroller programming and Python with the FastAPI framework for backend development. SQLite is used as the local database for storing access logs. The hardware components include an Arduino UNO, RFID reader (RC522) as the sensing unit, servo motor for gate actuation, LEDs, buzzer, and LED display for user feedback. A simple 3D printed enclosure and gate structure will be used to simulate a physical security barrier.

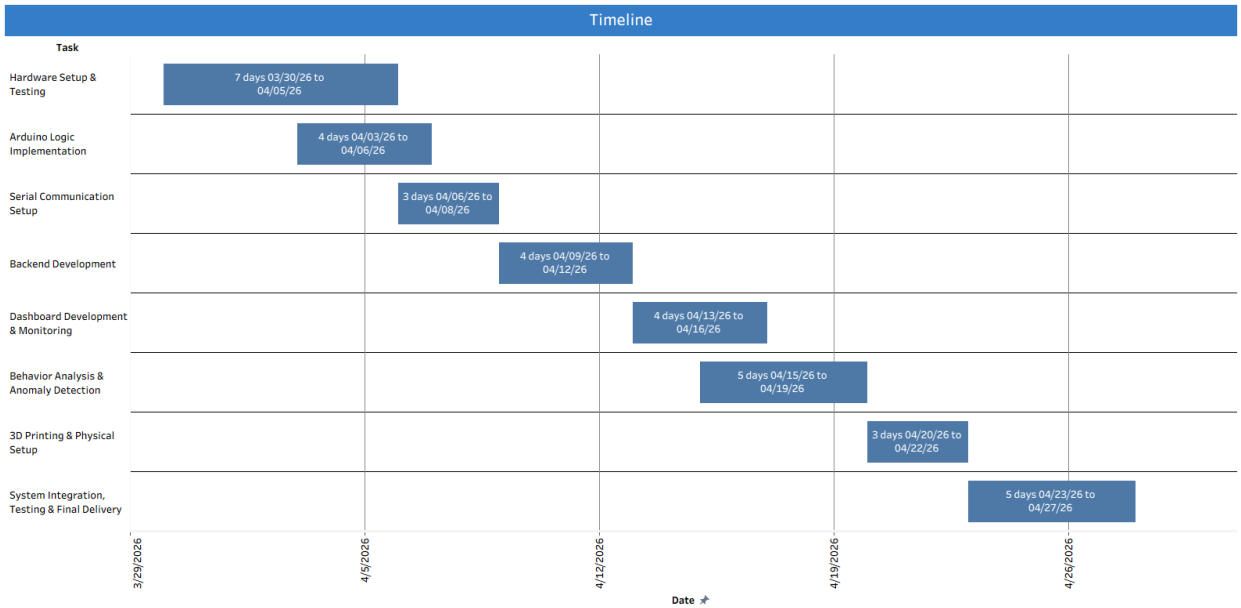
At the edge level, the Arduino continuously scans RFID cards and reads their unique identifiers (UIDs). The UID is compared against a predefined list of authorized users. If valid, the servo motor opens the gate, and the green LED and display indicate access granted. If invalid, access is denied, and the red LED and buzzer are activated. This ensures low-latency decision making without relying on external systems. Each access event is transmitted to a computer using USB serial communication (UART), which serves as a lightweight edge-to-backend communication method. The system can be extended to wireless technologies such as WiFi or MQTT in future deployments. The backend, developed using FastAPI,

receives the data and stores it in a SQLite database with UID, timestamp, access status, and entry/exit state. A web-based dashboard displays real-time activity and recent logs for monitoring.

Basic analytics are applied to identify usage patterns such as entry time and frequency. A simple anomaly detection approach, using rule-based logic or a lightweight model such as Isolation Forest, is used to detect unusual access behavior. The system will be developed in stages, starting with hardware integration and gate control logic, followed by communication setup, backend and database implementation, dashboard development, and finally behavior analysis, system integration, and testing.

4. Plan

The project will be completed in the following stages:



5. Delivery

The final project deliverables will include:

- A working hardware prototype of the security gate system
- A live demonstration of RFID-based access control
- A functional dashboard showing real-time and historical data
- A written report explaining system design and implementation
- Presentation slides summarizing the project

6. Conclusion

This project presents a practical implementation of an IoT-based access control system enhanced with monitoring and basic intelligence. By combining hardware-level authentication with backend data processing, the system goes beyond simple gate control and enables security tracking and behavior analysis. The project remains focused on core IoT concepts while demonstrating how additional software layers can improve system capabilities. This approach reflects real-world smart security systems and provides valuable experience in integrating embedded systems with data-driven applications.